

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Size,
Color

Behavior

Discount, Promo, Frequency, Rating,
Shipping

Data Preparation in Python

01

Load & Explore

Imported via pandas, used `df.info()` and `describe()`

02

Clean Missing Data

Imputed Review Rating using median per category

03

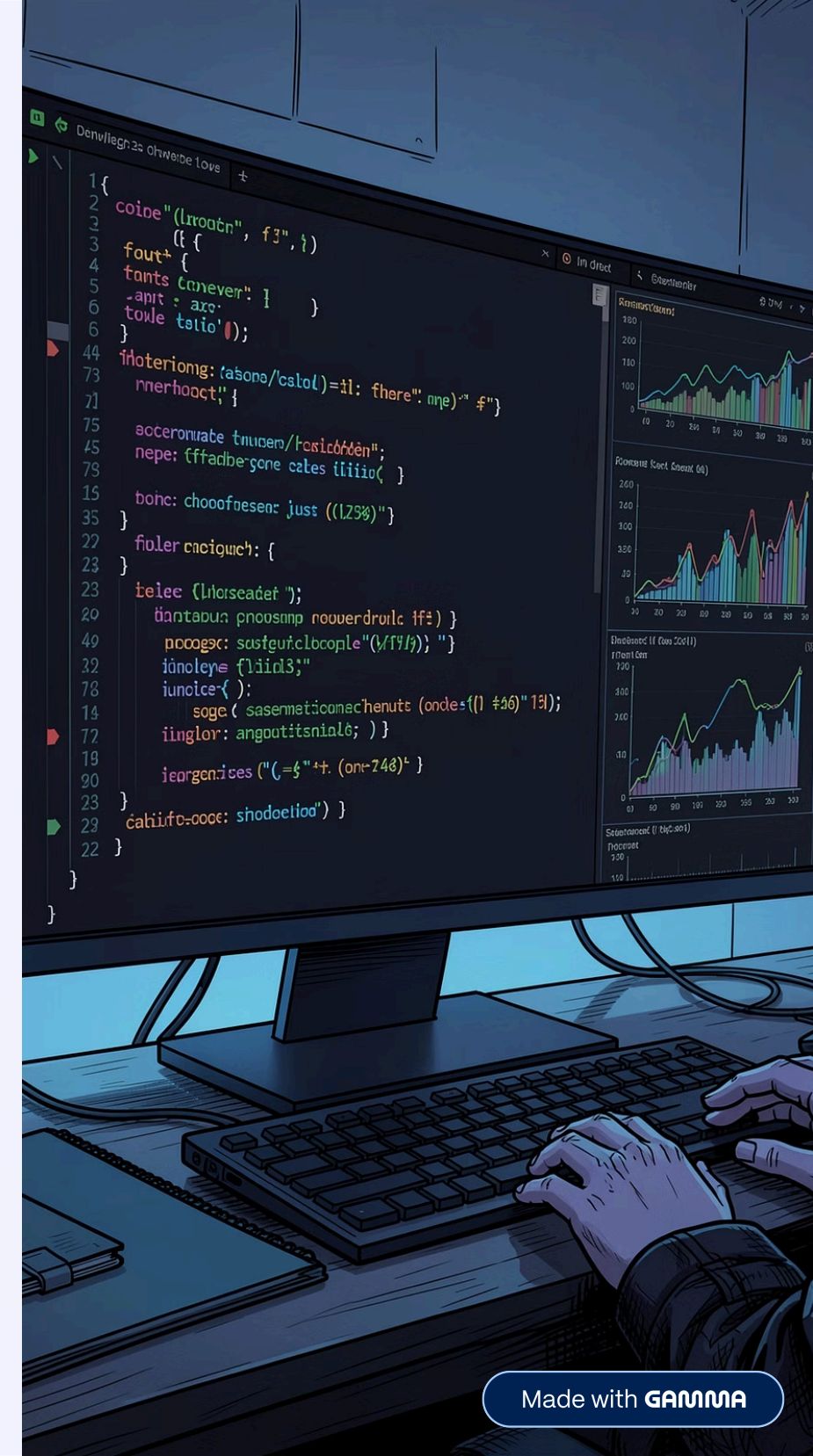
Feature Engineering

Created `age_group` and `purchase_frequency_days`

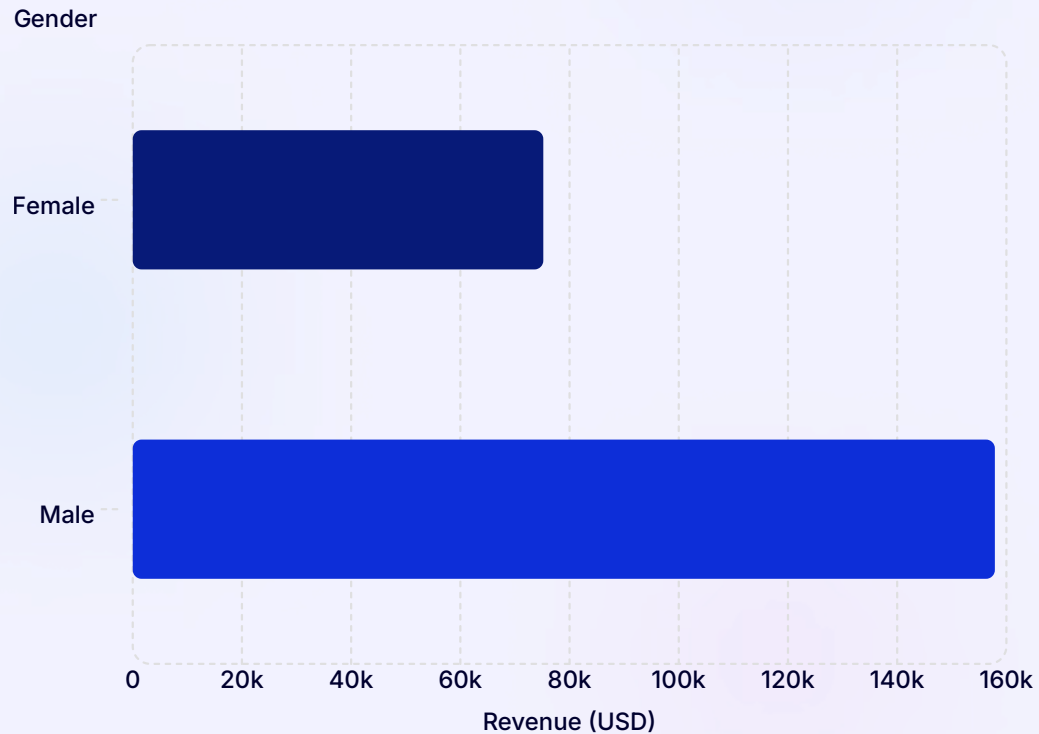
04

Load to PostgreSQL

Dropped redundant `promo_code_used`, connected to DB



Revenue by Gender



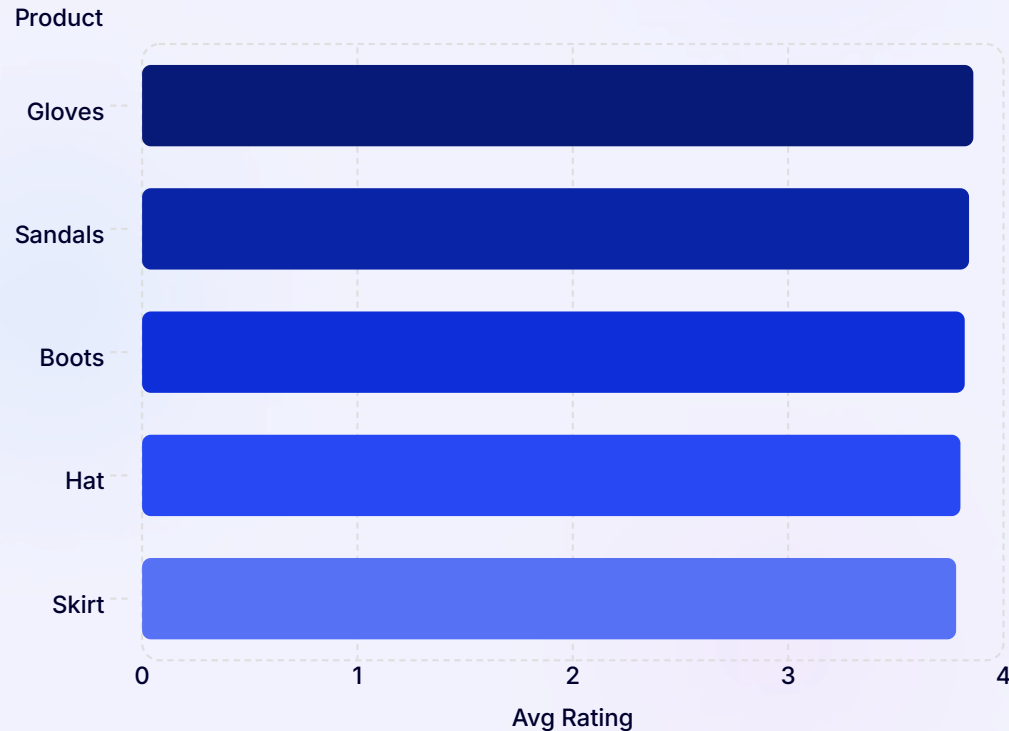
Key Insight

Male customers generated **2× more revenue** than female customers – \$157,890 vs. \$75,191.

- Male shoppers also represent 68% of total transactions (2,652 of 3,900).

Top Products & Discount Behavior

Top 5 Products by Rating



Most Discount-Dependent Products

Hat

50% discounted purchases

Sneakers

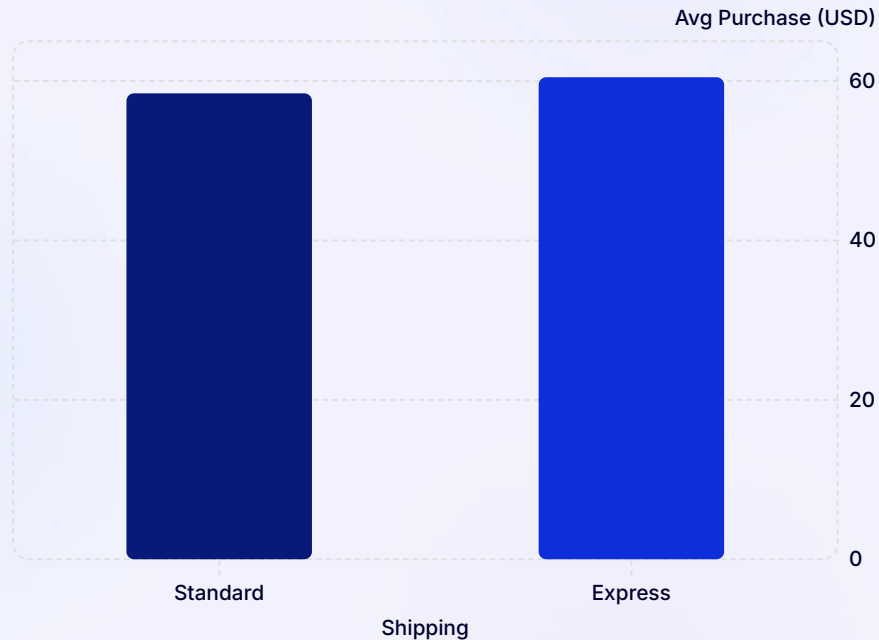
49.66% discounted

Coat

49.07% discounted

Shipping & Subscription Insights

Avg Spend by Shipping Type



Subscribers vs. Non-Subscribers

Repeat buyers (>5 purchases): **958 subscribers** vs. **2,518 non-subscribers**

Subscribers (27%)

1,053 customers · \$62,645 revenue

Non-Subscribers (73%)

2,847 customers · \$170,436 revenue

Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

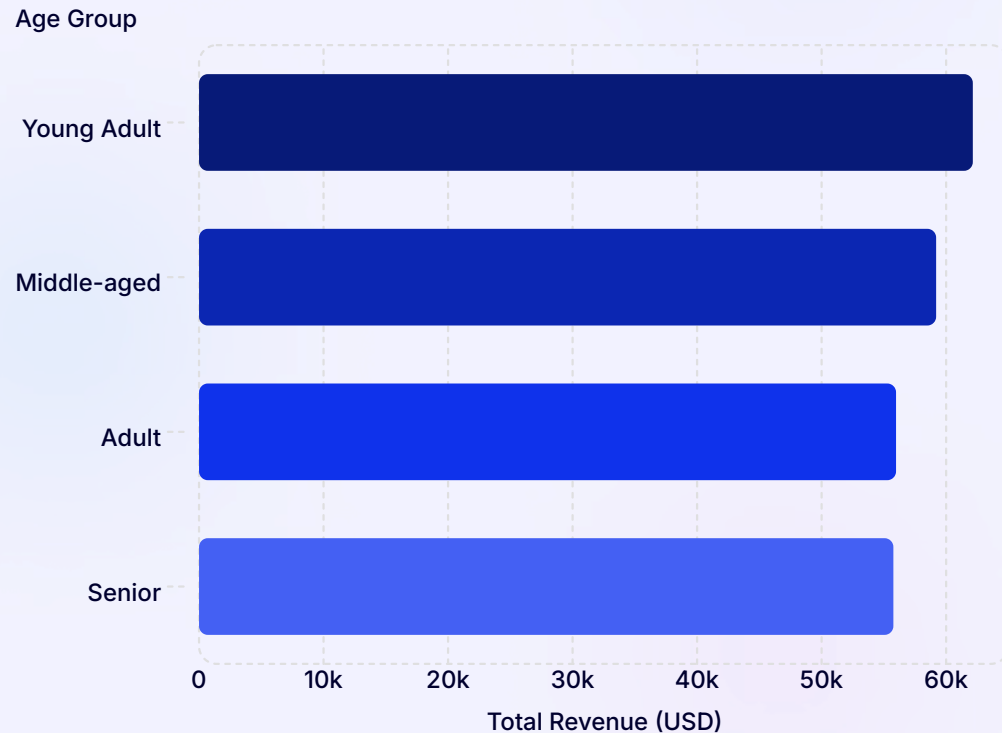
New

83 customers

839 high-spending customers used discounts yet spent **above average** purchase amount.



Revenue by Age Group



Top 3 Products per Category

Clothing

Blouse · Pants · Shirt

Footwear

Sandals · Shoes ·
Sneakers

Outerwear

Jacket · Coat

Accessories

Jewelry · Sunglasses ·
Belt

Power BI Dashboard



942 Customers

Avg purchase: AED 59.42



3.73 Avg Rating

Across all products



Interactive Filters

Subscription, Gender, Category



Visualizations

Revenue & Sales by Category and Age Group



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward Loyal segment



Review Discount Policy

Balance sales lifts with margin control



Targeted Marketing

Focus on high-revenue age groups and express-shipping users